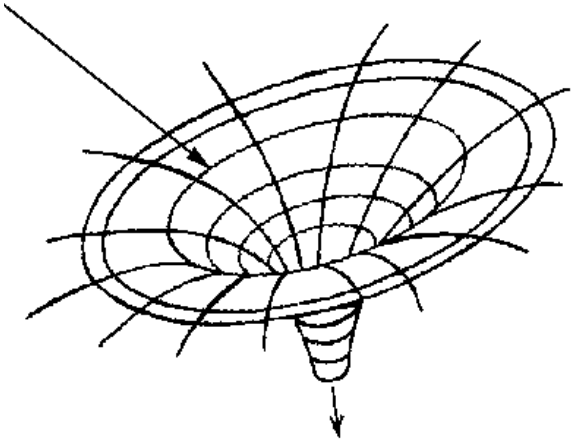
You Did Not Just Read This

or did you?

The Reconciliation of Quantum Theory and General Relativity

No Strings Attached



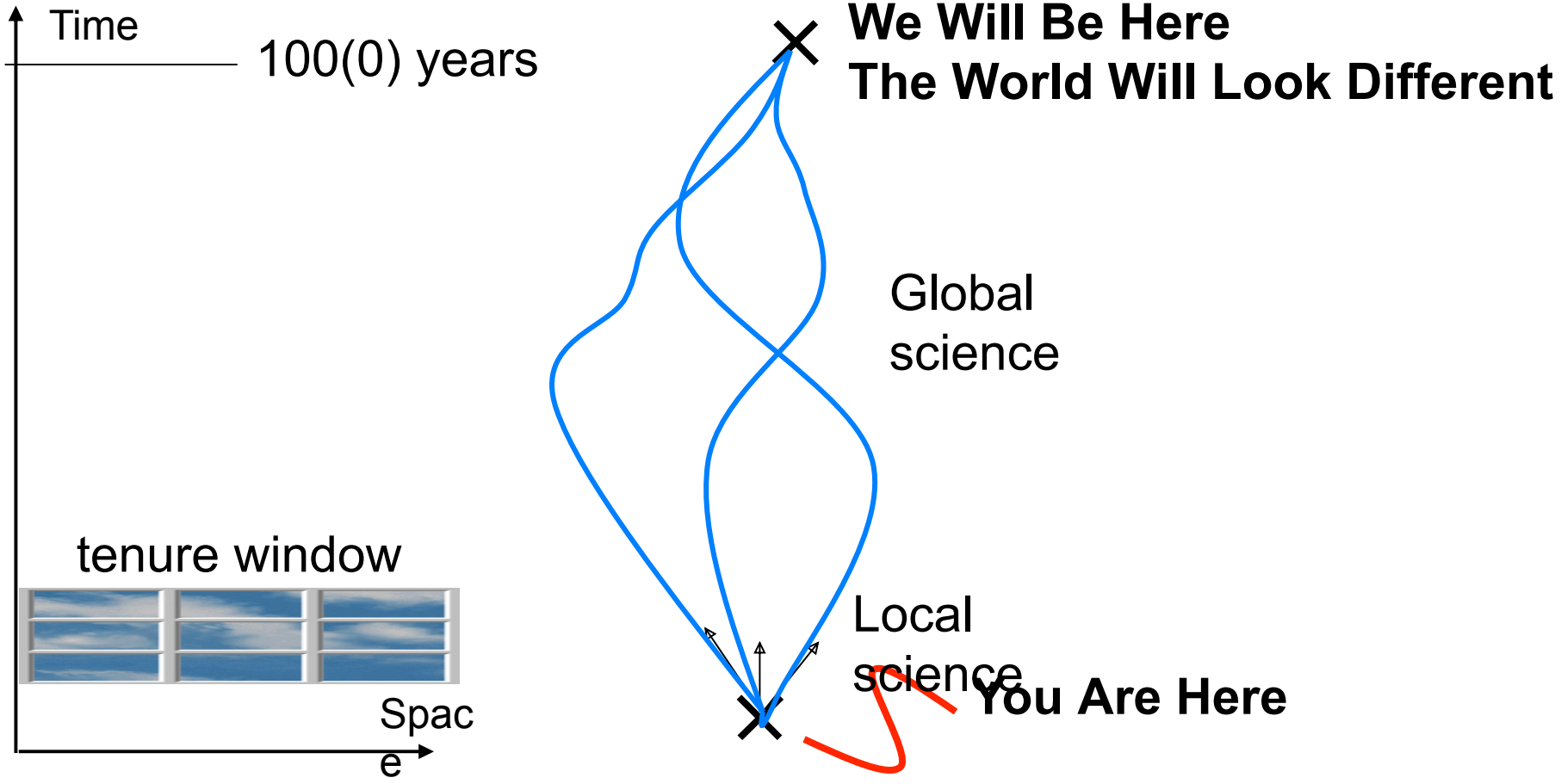


Dave Bacon

Santa Fe

Possible Paths

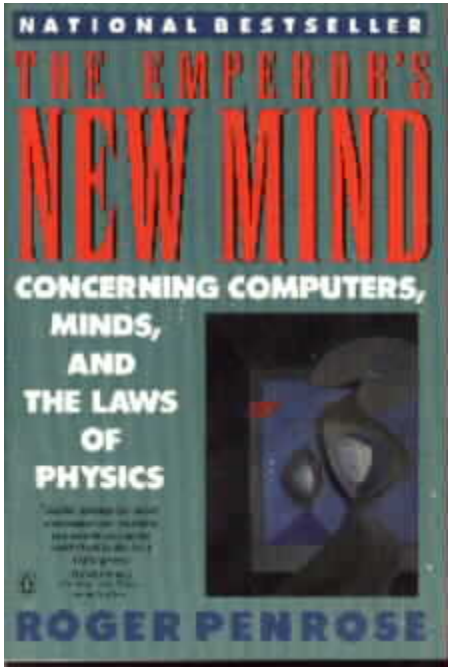
They flutter behind you your possible pasts,
Some brighteyed and crazy, some frightened and lost.
A warning to anyone still in command
Of their possible future, to take care. – Pink Floyd, Your Possible Pasts



Book Deal

This talk morphed into:

"If I were going to write a book speculating about the future of quantum theory and general relativity"



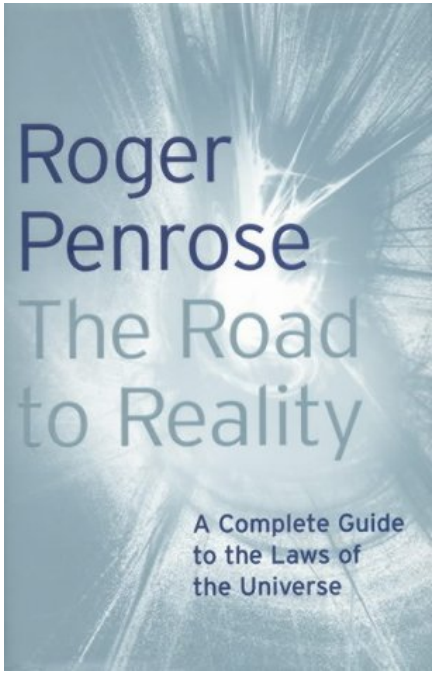
Lots of history

Lots of history of speculative ideas

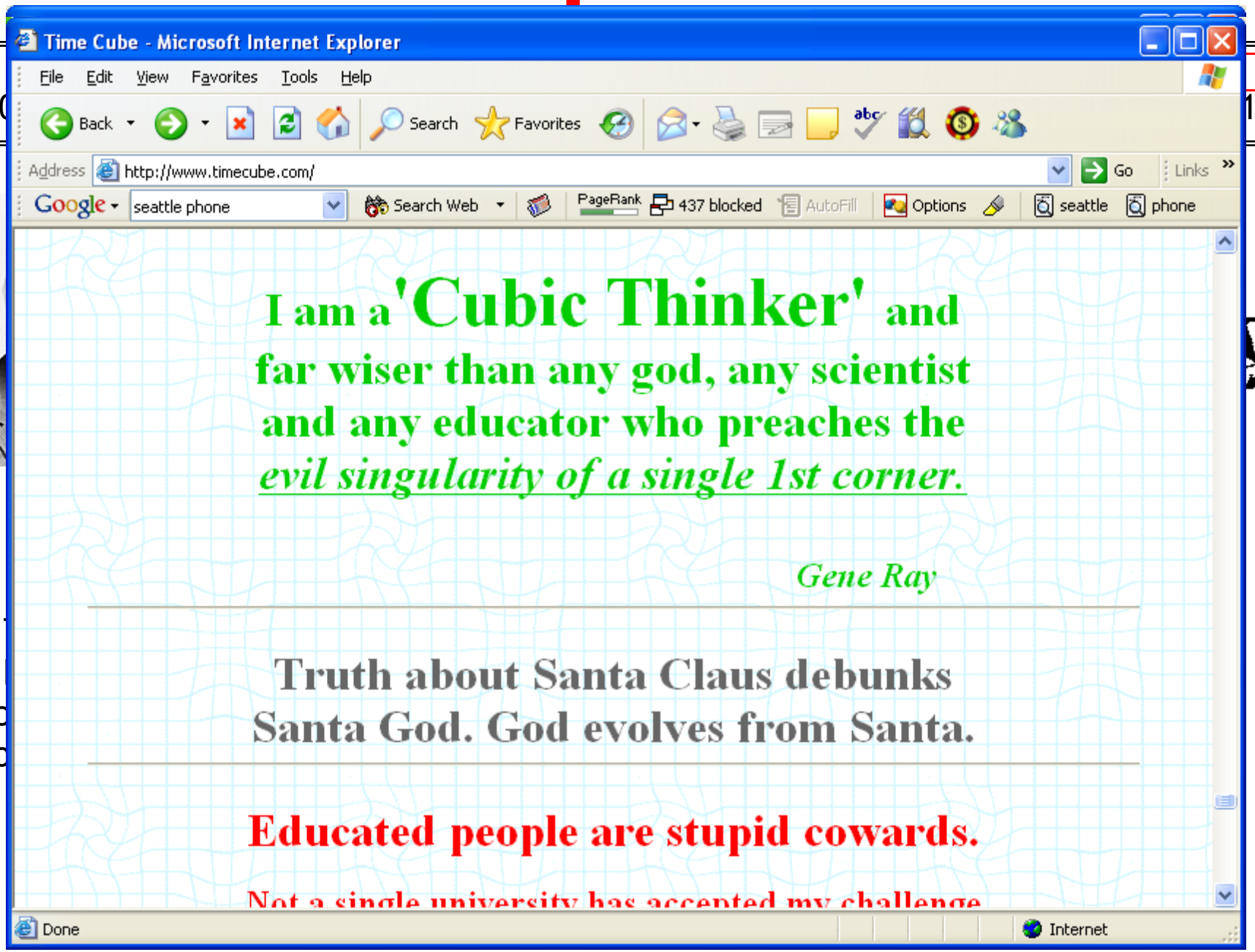
Speculative ideas

Who is my audience?

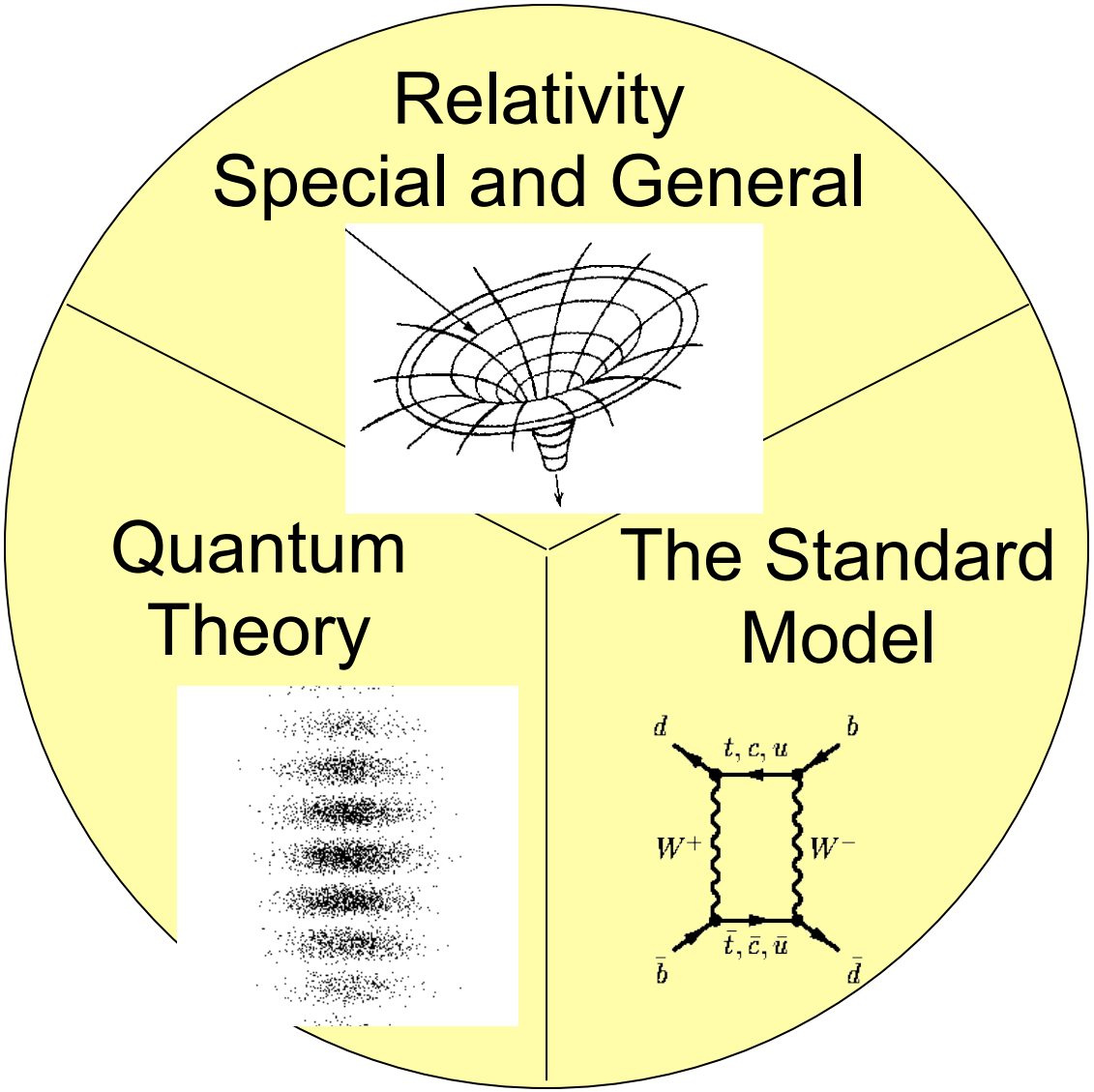
Healthy(?) skepticism



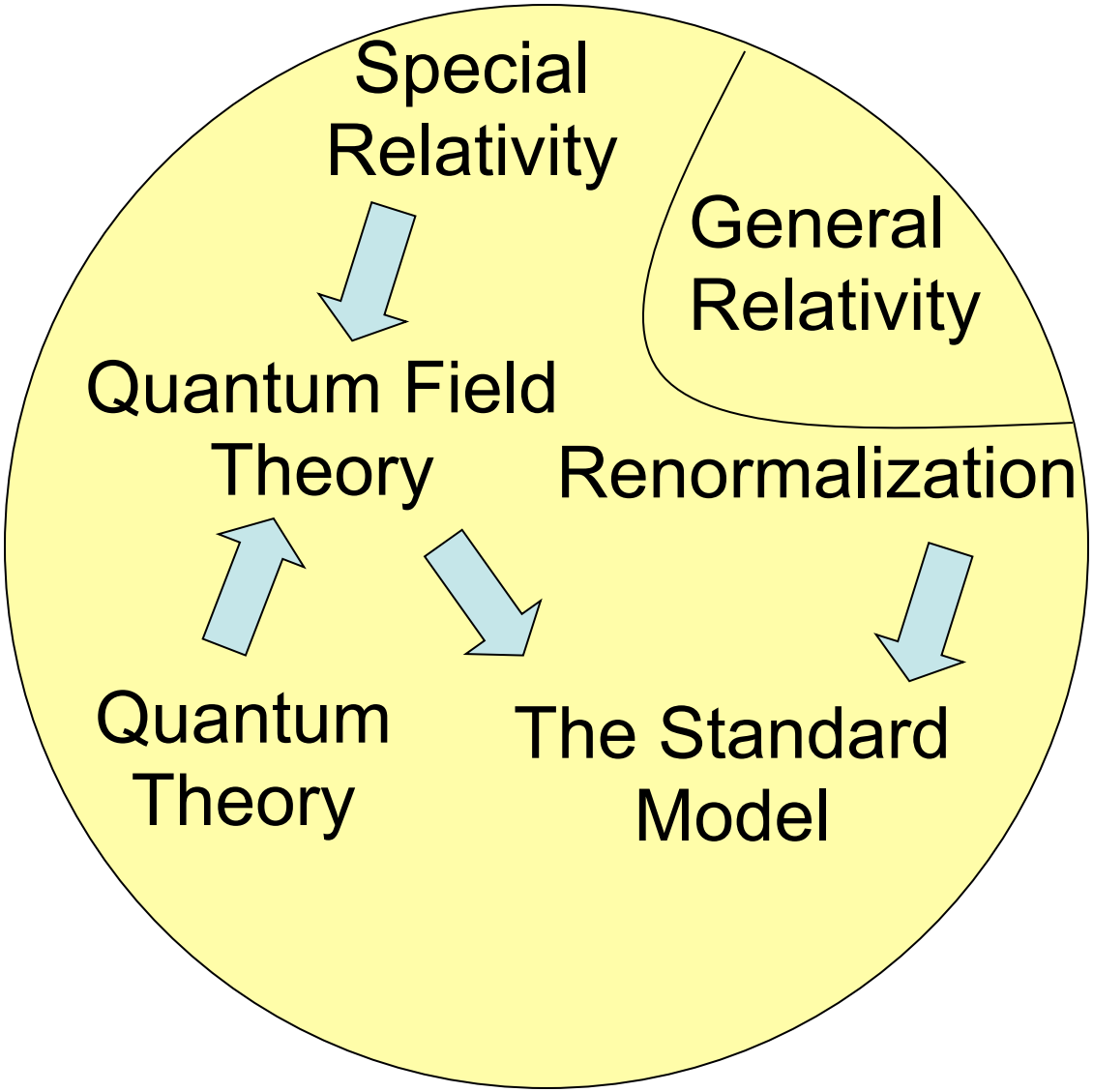
Crackpot Meter

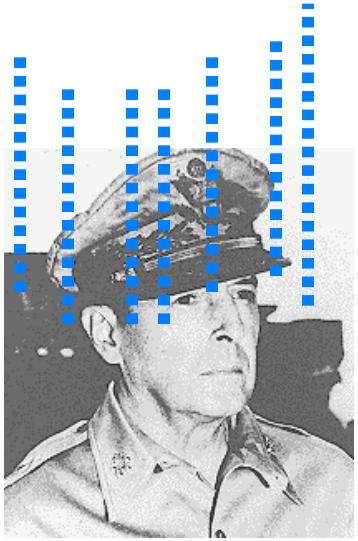


Three Revolutions of Fundamental Modern Physics



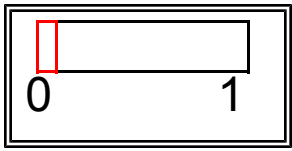
Three Revolutions of Fundamental Modern Physics







The Failure

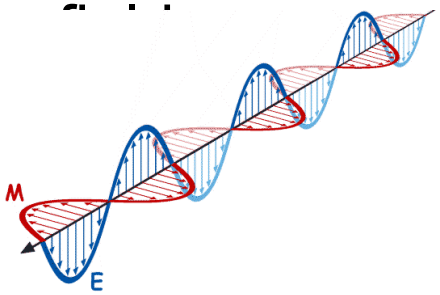


Quantum Field Theory

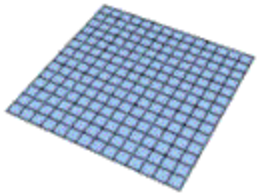
General Relativity

dynamic variables

particle



metri



$$\phi(x)\ \psi_\mu(x)\ A_\mu(x)\ G^\alpha_\mu(x)$$

$$g\mu,\nu$$

defined over some space-time

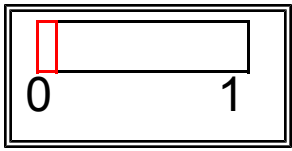
space-time itself!

the game called renormalization

works

fails

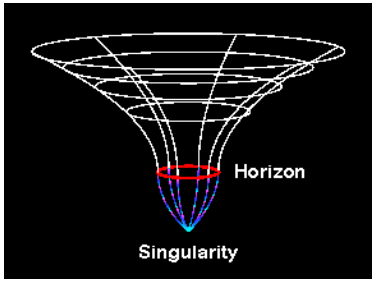
Planck Length



General Relativity

Blackholes: $ds^2 = \left(1 - \frac{2GM}{rc^2}\right)^{-1} dr^2 + r^2 d\Omega^2 - \left(1 - \frac{2GM}{rc^2}\right)^{-1} c^2 dt^2$

If we cram mass inside $r < r_s = \frac{2GM}{c^2}$
we create a blackhole



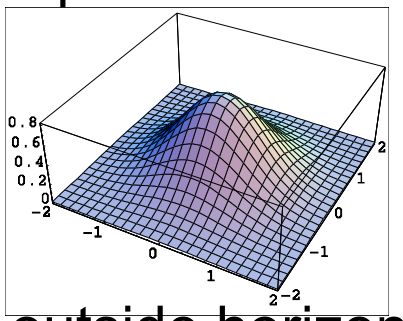
Any mass blackhole possible

Quantum Field Theory

$\Delta x \Delta p \geq \frac{\hbar}{2}$ localize to diameter d → large momentum possible

large momentum mc → particle creation

Compton length: $l_c = \frac{\hbar}{Mc}$



Black holes of small mass such that Compton length is outside horizon?

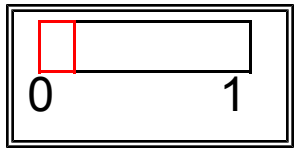
$M_P = \sqrt{\frac{\hbar c}{G}} \approx 10^{-8}$ kg

Planck Mass

$L_P = \sqrt{\frac{G\hbar}{c^3}} \approx 10^{-35}$ m

Planck Length

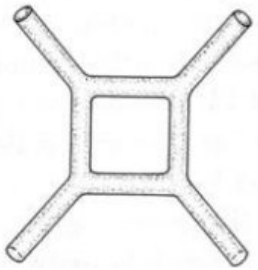
The Beaten Path



String Theory:

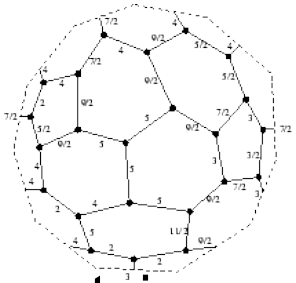
- quantum theory of strings in high dimensional spaces
- graviton (spin 2) naturally arises
- hope to incorporate all of Standard Model

Big Problem: no string universe looks like our own (the landscape problem), what is M-theory?



Loop Quantum Gravity:

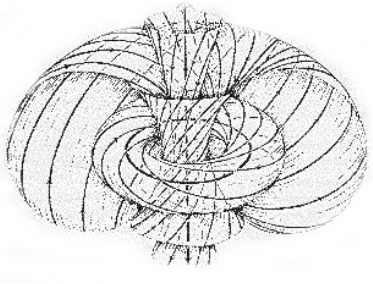
- non-canonical quantization of Ashtekar's variables
- gravity only
- non-perturbative



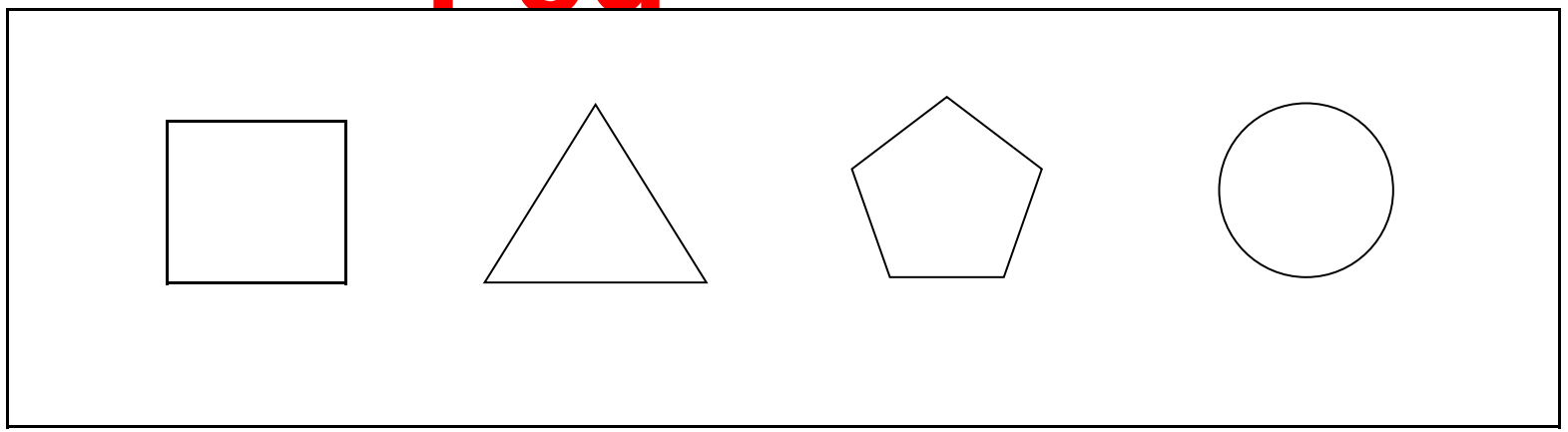
Big Problem: emergence of classical general relativity uncertain, proper dynamics not established, can we calculate anything?

Others:

Euclidean quantum gravity, Twistors, Causal sets, Dynamical triangulation, Supergravity, Causal dynamical triangulation, ….

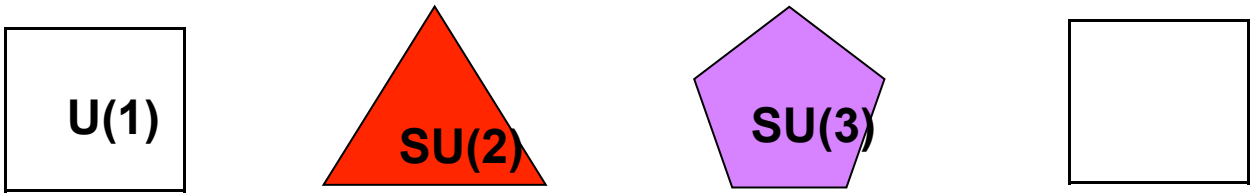


Square Peg



second quantization

renormalization



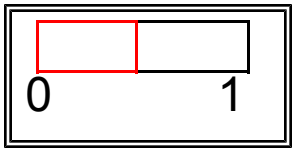
Electromagnetic Force

Weak Forc

Strong Force

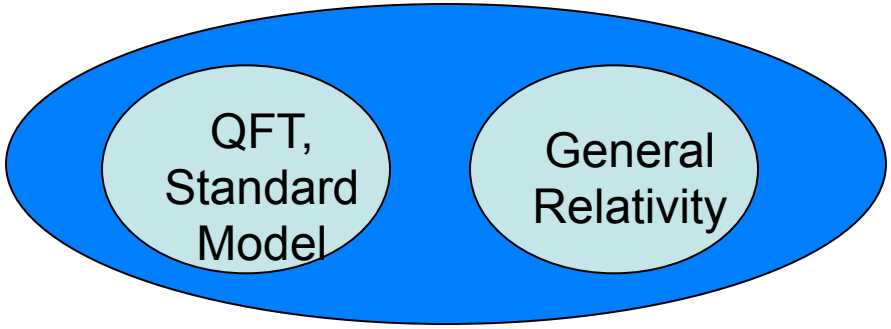
Gravitational Force

The Unbeaten Path



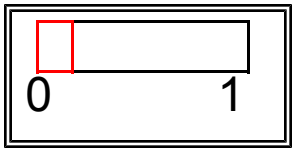
Instead of trying to quantize gravity, a “possible path” is to

> Derive quantum (field) theory from a deeper theory of spacetime which includes general relativity as a limit.



1. Motivation. Defining properties of quantum theory.
2. Argument by anecdote.
3. The possible path.
4. Technological implications.

Quantum Theory

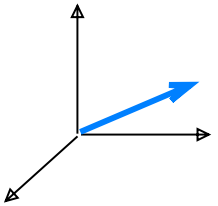


"We all know how to use [quantum mechanics] and how to apply it to problems; and so we have learned to live with the fact that nobody can understand it."

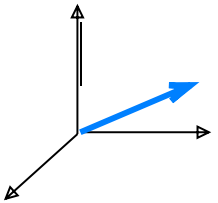
- Murray Gell-Mann

Postulates of Quantum Theory

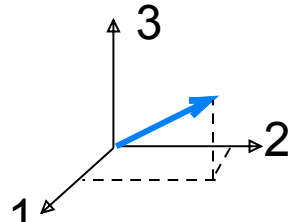
1. The state of a system is described by a wave function.
   The wave function is an element of a Hilbert space (think: a vector).



2. Evolution of the state of the quantum system is described by a unitary operator acting on the wave function (think: rotating vector)

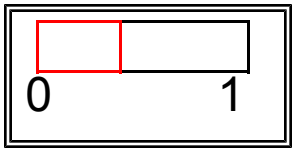


3. From the wave function we can calculate the probability of measurement outcomes by examining the overlap squared of the wave function with a set of vectors describing the measurement. (think: projecting vector)



4. State space for different degrees of freedom combine as a tensor product. (think: d1+d2 dim vector as concatenation of d1 dim vector and d2 dim vector)

Quantum Non-Mysteries



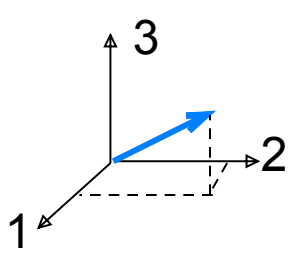
The wave function and reality

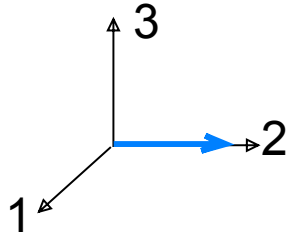
we have no right to demand a connection between the wave function and elements of reality

Kochen-Specker Theorem: those measurement directions do not correspond to elements of reality.

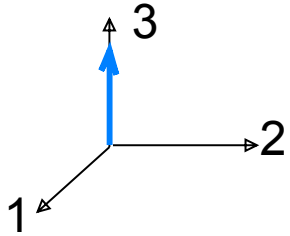
Silliness: “intrinsic quantum uncertainty”

quantum theory *contains* determinism





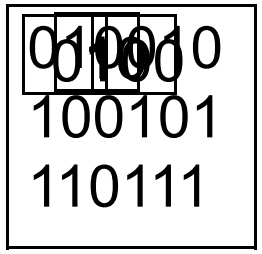
“100% 2”



“100% 3”

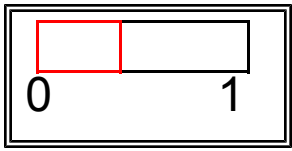
KS: no way to assign “100% 2” with an element of reality “2”.

information describing system



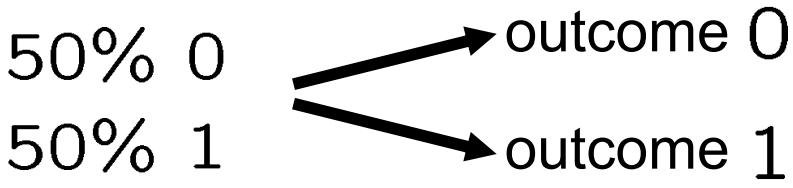
measurement device property: B

Quantum Non-Mysteries



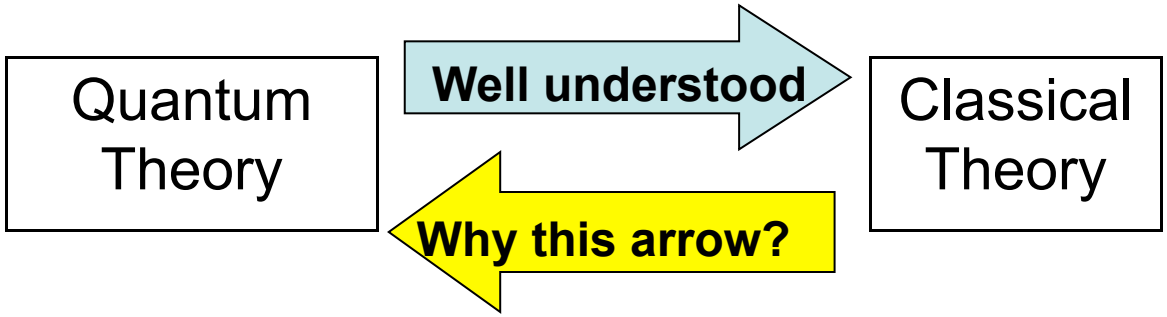
"collapse" of the wave function

any theory which only predicts probabilities will have "collapse"



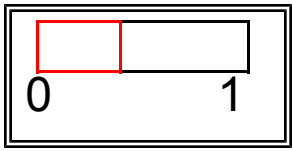
Conflict between classical physics and quantum physics

the emergence of classical physics from quantum theory is not "totally" solved, but we sure know a lot about how this occurs



We should not demand "classicality" in a theory which explains the origin of quantum theory

Quantum Mysteries

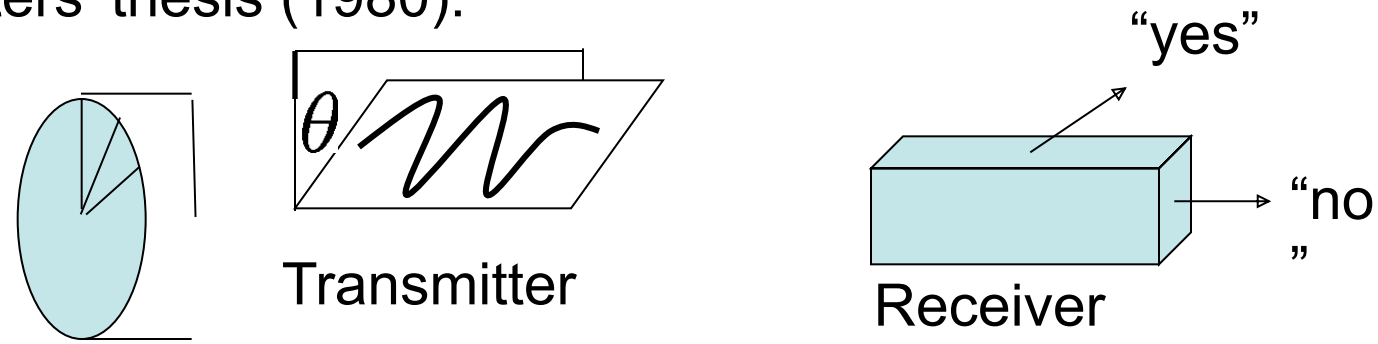


Why Hilbert space? Why the quantum probability law?

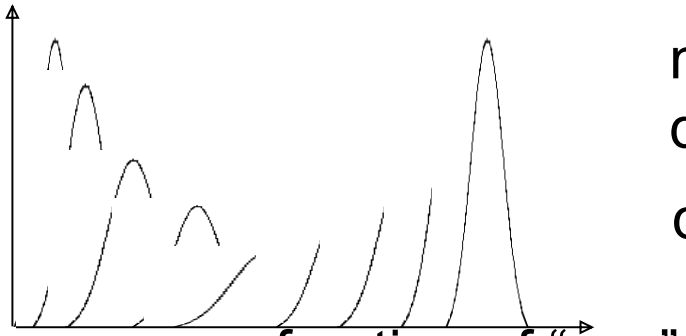
Quantum theory has been very universal:
all of the standard model obeys quantum theory

How do we separate out the information content of quantum theory from deeper physics? (The Fuchs Program)

Wootters' thesis (1980):



600 yes, 400 no → root mean square deviation ≈15.5.

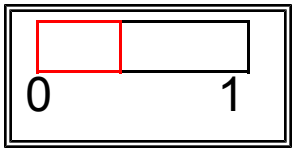


maximize number of distinguishable distinguishable angles

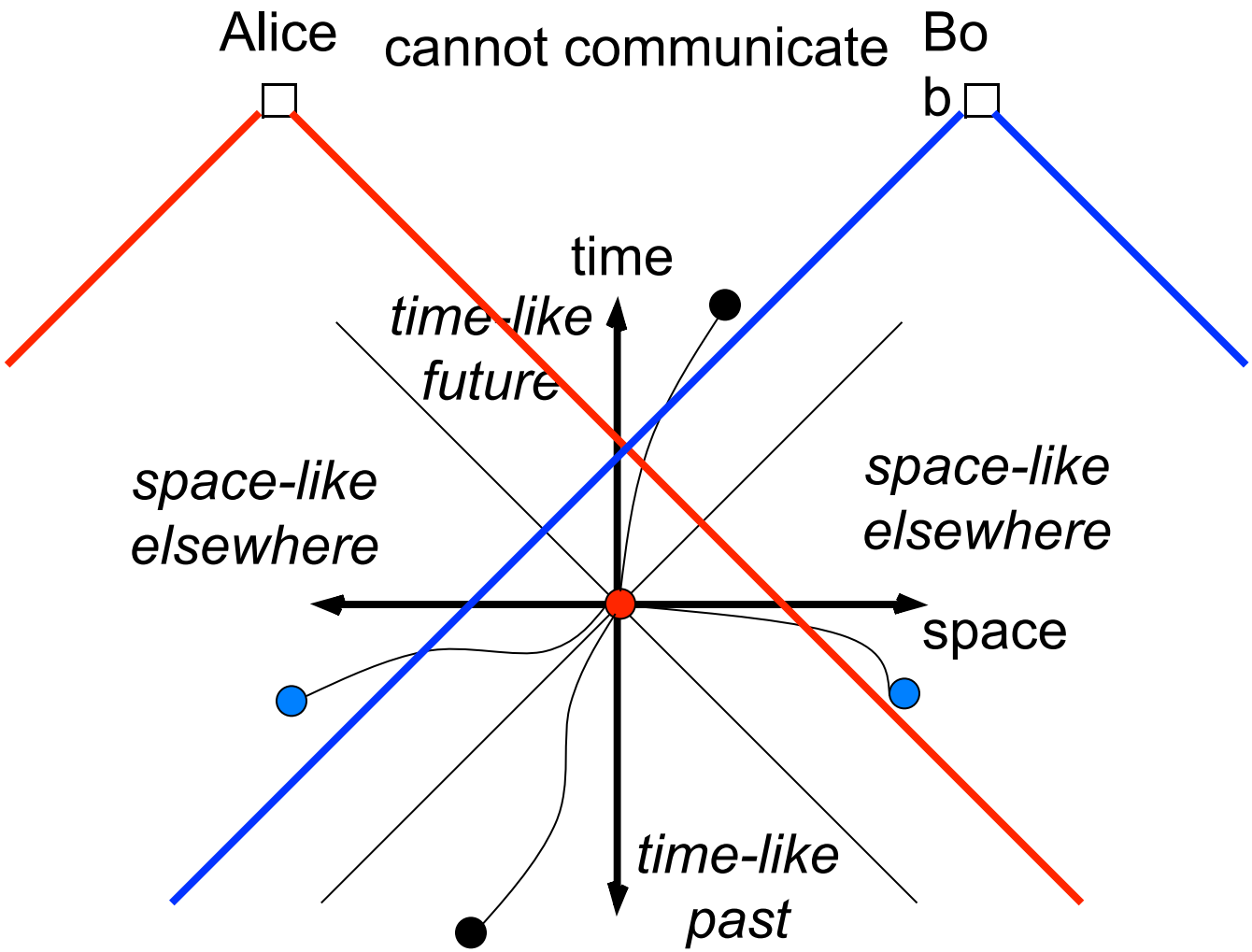
quantum probability law:

$$P(\theta) = \cos^2(n\theta)$$

Quantum Mysteries

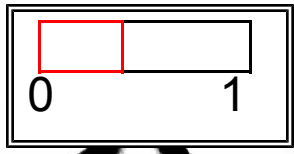


Information and Locality

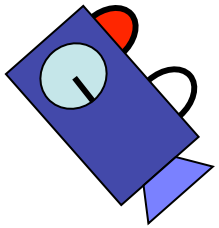


Special relativity gives rise to a causal structure for the transmission of classical information.

Compar



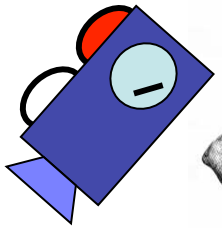




ALICE

two quantum subsystems

e





BOB

2. Separate

1. Interact and entangle

3. Choose measurement

4. Measure

5. Get outcome

3. Choose measurement

4. Measure

5. Get outcome



ALICE

1. Share classical correlations

2. Choose measurement

2. Choose measurement

data

03XSF
W3331

data

03XSF
GF111

3. Produce measurement outcome

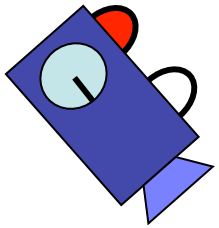
3. Produce measurement outcome

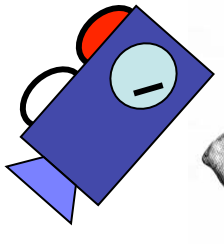


BOB

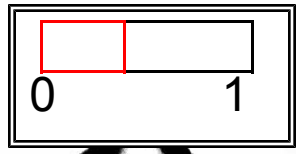
An Experiment







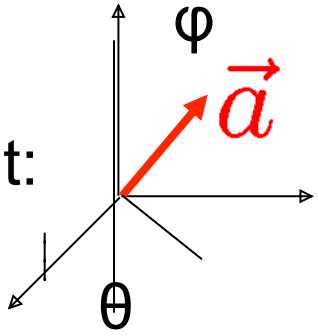




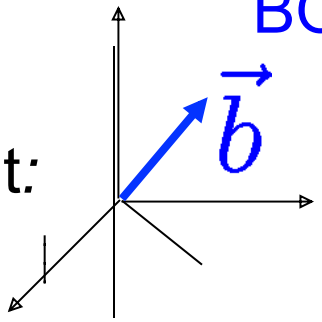
ALICE

BOB

Measurement:



Measurement:



Outcome : $A \in \{+1,-1\}$

Outcome : $B \in \{+1,-1\}$

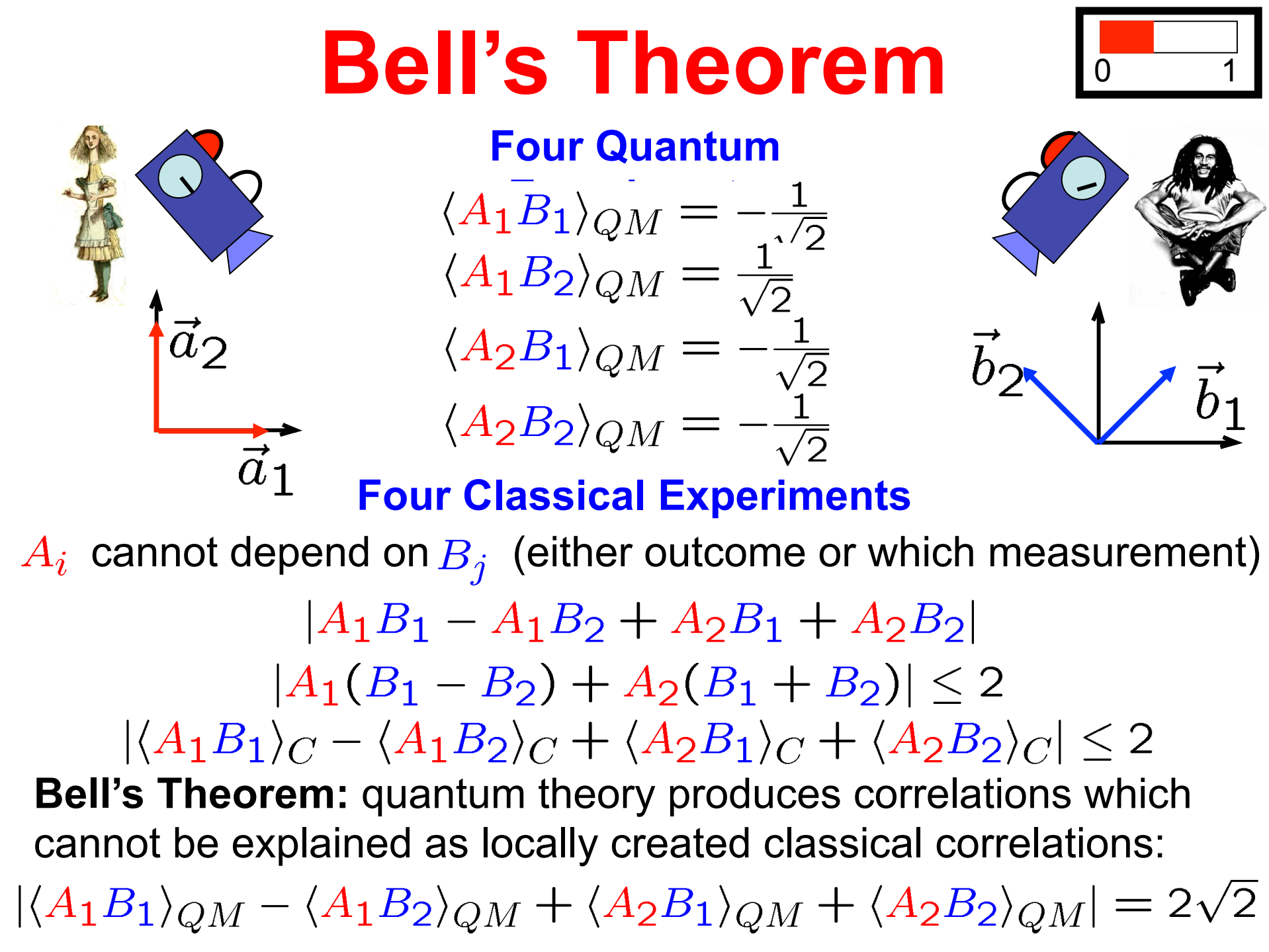
Quantum Experiment Results:

$$\Pr\left[A, B|\vec{a}, \vec{b}\right] = \frac{1}{4}\left[1 - AB\vec{a}\cdot\vec{b}\right]$$

$$\langle A\rangle = \langle B\rangle = 0$$

$$\langle AB\rangle = -\vec{a}\cdot\vec{b}$$

Bell's Theorem



Four Quantum

$$\langle A_1 B_1 \rangle_{QM} = -\frac{1}{\sqrt{2}}$$
$$\langle A_1 B_2 \rangle_{QM} = \frac{1}{\sqrt{2}}$$
$$\langle A_2 B_1 \rangle_{QM} = -\frac{1}{\sqrt{2}}$$
$$\langle A_2 B_2 \rangle_{QM} = -\frac{1}{\sqrt{2}}$$

Four Classical Experiments

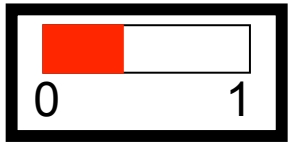
A_i cannot depend on B_j (either outcome or which measurement)

$$|A_1 B_1 - A_1 B_2 + A_2 B_1 + A_2 B_2|$$
$$|A_1 (B_1 - B_2) + A_2 (B_1 + B_2)| \leq 2$$
$$|\langle A_1 B_1 \rangle_C - \langle A_1 B_2 \rangle_C + \langle A_2 B_1 \rangle_C + \langle A_2 B_2 \rangle_C| \leq 2$$

Bell's Theorem: quantum theory produces correlations which cannot be explained as locally created classical correlations:

$$|\langle A_1 B_1 \rangle_{QM} - \langle A_1 B_2 \rangle_{QM} + \langle A_2 B_1 \rangle_{QM} + \langle A_2 B_2 \rangle_{QM}| = 2\sqrt{2}$$

Quantum Mysteries



Bell's Theorem:

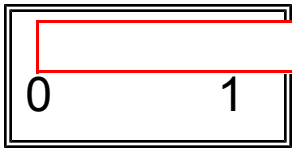
Assumptions:

1. classical information describing quantum system (weak realism)
   (note: no assumption that we can measure this information, KS)
2. no cosmic conspiracy
3. causal structure of space-time

Comments:

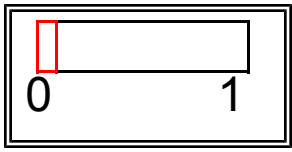
Bell's theorem is stronger than just ruling out "classical physics" explanations of quantum theory.

No faster than light communication. No spooky action at a distance.
No faster than light communication. No spooky action at a distance.
No faster than light communication. No spooky action at a distance.
No faster than light communication. No spooky action at a distance.
No faster than light communication. No spooky action at a distance.



Mysterious Extra Slide

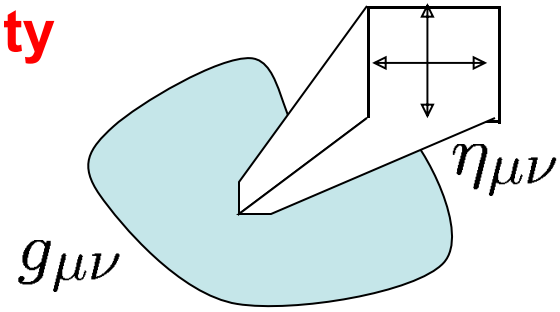
General Relativity



"Matter tells space-time how to bend and space-time tells matter how to move"
-John Wheeler

Equivalence principle: all physical laws are the same in all inertial and freely falling frames of reference (gravitational mass = inertial mass)

Postulates of General Relativity



1. Space-time is a four-dimensional pseudo-Riemannian manifold whose metric signature is (-1,+1,+1,+1).

2. Matter/energy is described by the stress energy tensor, a two index tensor describing both the static and dynamic properties. $T_{\mu\nu}$

3. Einstein field equations.

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi T_{\mu\nu}$$

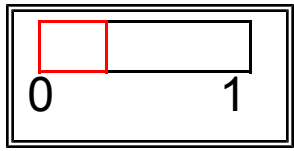
Einstein tensor

Cosmological constant

Nonlinear partial differential equations for metric.

(4). Sometimes we also add that test particles travel along geodesics (but we could instead talk about singularities, limit of small mass)

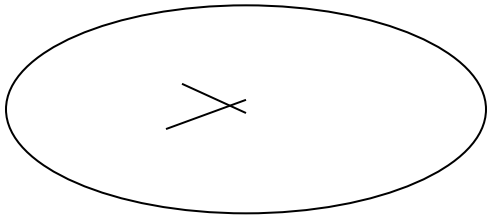
General Relativity's Muteness



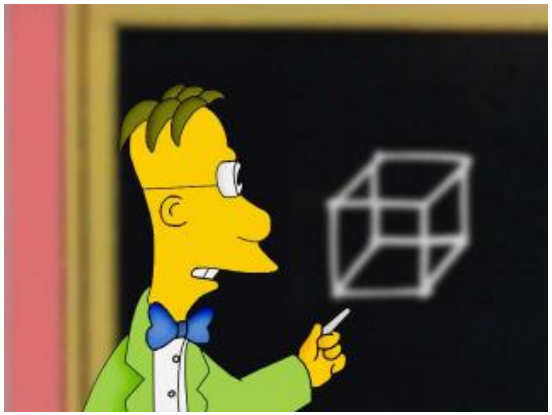
General Relativity's incompleteness about spacetime:



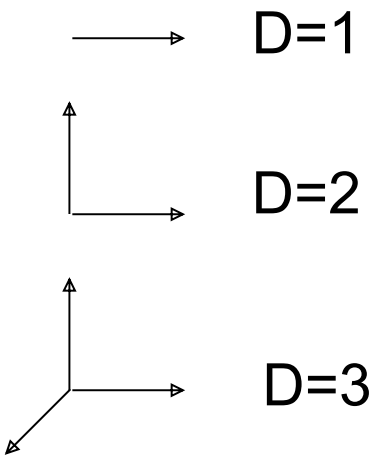
Topology



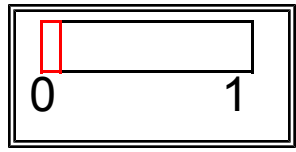
and

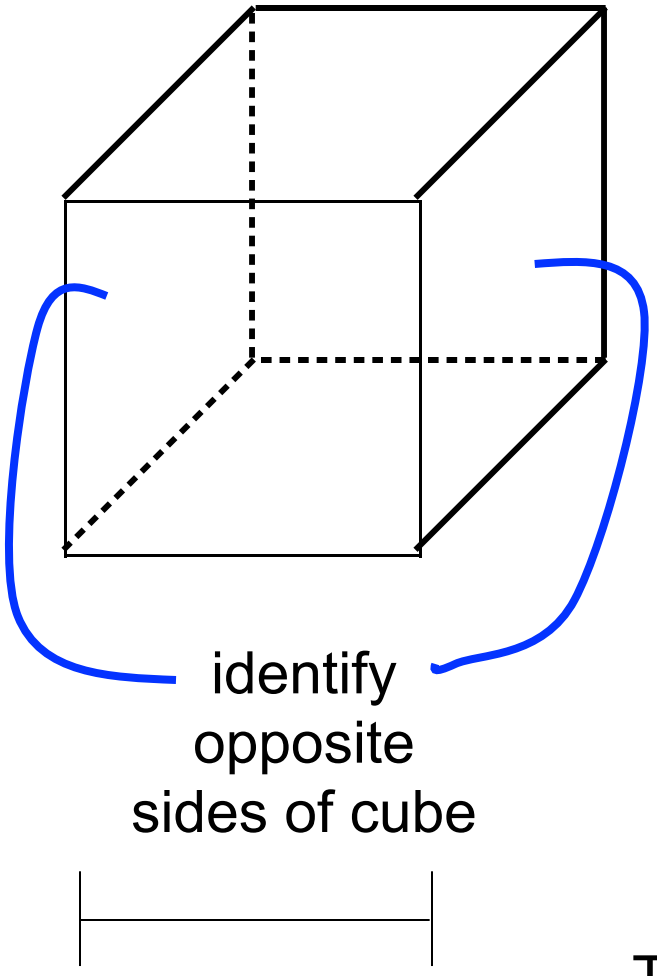


Dimension



Twins in Torusland



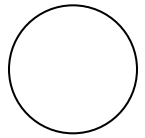


identify
opposite
sides of cube

preferred reference frame

Topology:

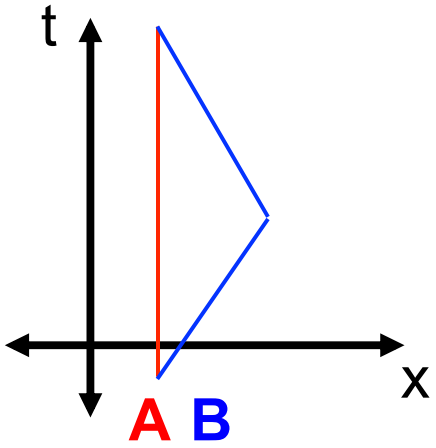
$$\mathbb{S}^3 \times \mathbb{R} \qquad \mathbb{S} :=$$



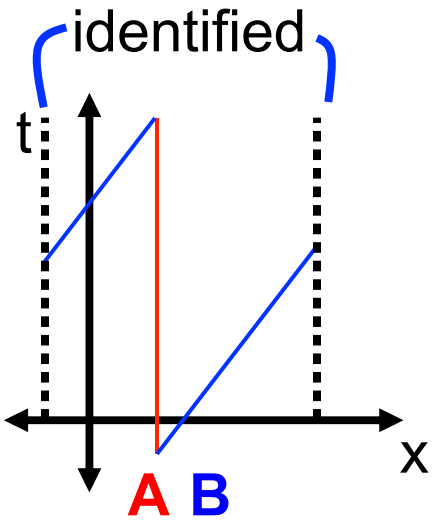
Metric: flat

$$\eta_{\mu\nu} = diag(-1, 1, 1, 1)$$

Twin paradox:

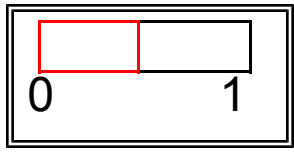


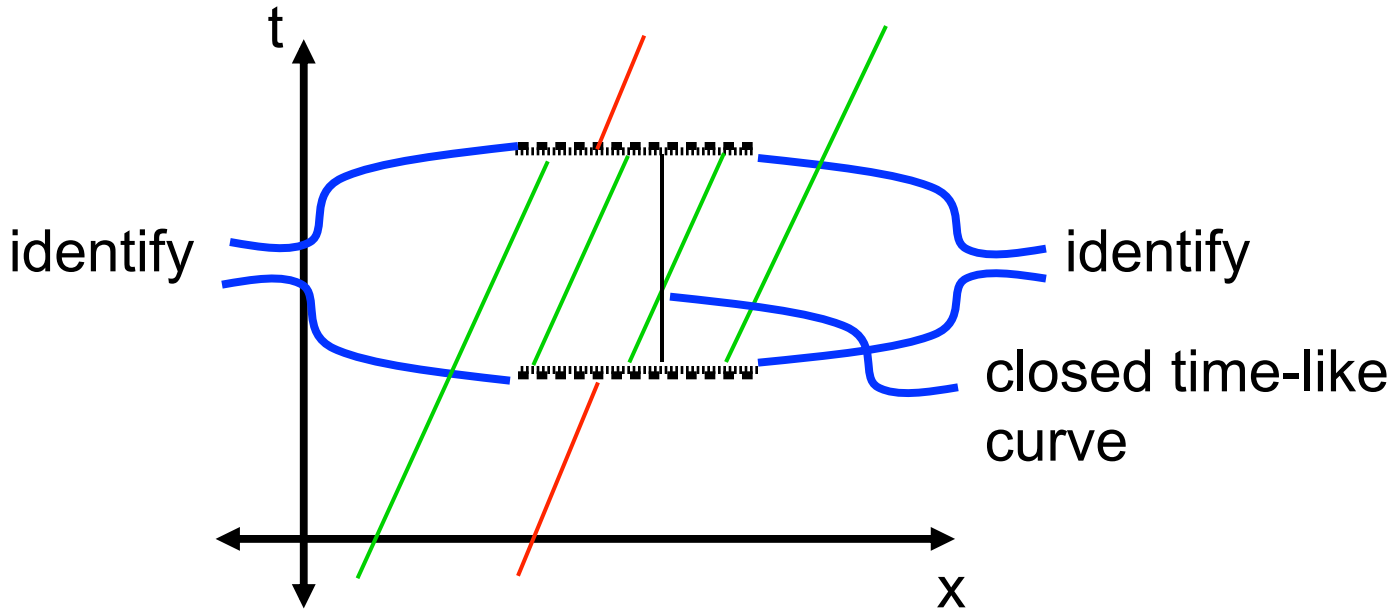
Time dilation implies
A is older than B



who is older?

Topology and Murder





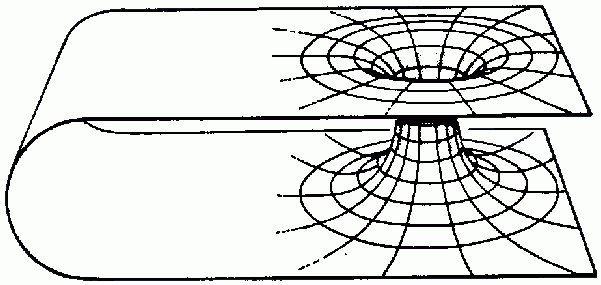
Chronology protection conjecture (Hawking):
The lord doth prohibit spacetimes with closed timelike curves.

Moral: Some topologies are harmless, some lead to severe headaches.

T. Brun, "Computers with closed timelike curves can solve hard problems efficiently," Foundations of Physics Letters 16, 245 (2003).
D. Bacon, "Quantum computation in the presence of closed timelike curves," Phys. Rev. A 70, 032309 (2004)
S. Aaronson, "NP-complete Problems and Physical Reality," SIGACT News Complexity Theory Column, March 2005.

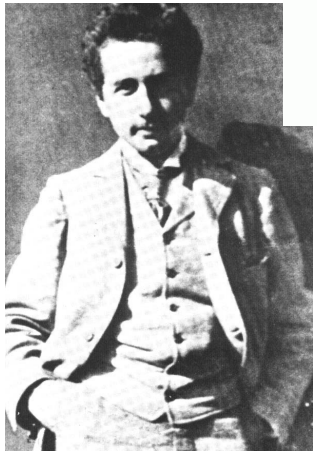


Old



n

A. Einstein and N. Rosen 1935



Einstein-Rosen Bridge
(a.k.a. “a wormhole”)

Paste together two
Schwarzschild throats

Basically
unstable.

Not possible to traverse wormhole before the throat pinches itself off.

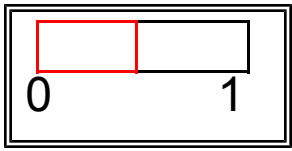
IN spite of its great success in various fields, the present theoretical physics is still far from being able to provide a unified foundation on which the theoretical treatment of all phenomena could be based. We have a general relativistic theory of macroscopic phenomena, which however has hitherto been unable to account for the atomic structure of matter and for quantum effects, and we have a quantum theory, which is able to account satisfactorily for a large number of atomic and quantum phenomena but which by its very nature is unsuited to the principle of relativity. Under these circumstances it does not seem superfluous to raise the question as to what extent the method of general relativity provides the possibility of accounting for atomic phenomena. It is to such a possibility that we wish to

Note: Einstein, Rosen paper publish two months after famous Einstein, Podolsky, Rosen paper. Can we deduce Einstein’s response to Bell’s theorem?

Crank checklist: references to Einstein



JAWs Says



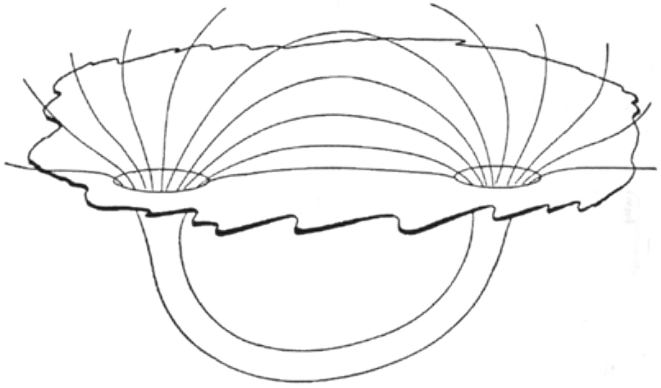
Gauss's Law

$$\nabla \cdot E = 4\pi\rho$$

point particles as singularities in the electric field

John Archibald Wheeler (1957)

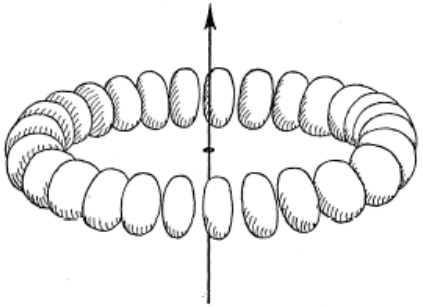
Charge without charge: source free Gauss's law on wormhole topology makes throats look like charged particles



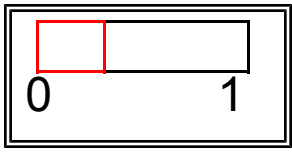
$$\nabla \cdot E = 0$$

Wheeler and Misner's attempts a using source free Maxwell equations, general relativity, plus topology to derive all of classical physics.

Random extra JAW cool thing: geon



Serious Point Particles



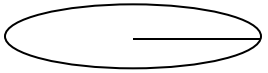
Spinning (J), charged (Q), massive (M) object: Kerr-Newman solution

$$ds^2 = -\frac{\Delta}{\rho^2}\left(dt - a\sin^2\theta d\phi\right)^2 + \frac{\sin^2\theta}{\rho^2}\left[(r^2+a^2)d\phi - a dt\right]^2 + \frac{\rho^2}{\Delta}dr^2 + \rho^2 d\theta^2$$

$$\rho^2 = r^2 + a^2\cos^2\theta \qquad a = \frac{J}{M} \qquad \Delta^2 = r^2 - 2Mr + a^2 + Q^2$$

Take seriously for elementary particles?

Solution has a naked (no event horizon) ring
Singularity with closed timelike curves!



$$r = a = \frac{J}{M}$$

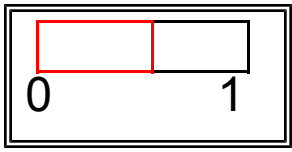
spin s particle: $r = \frac{s\hbar}{Mc} = s l_C$

$$r = \frac{J}{\hbar}\frac{M_P}{M} l_P = \frac{J}{Mc}$$

Naked singularity of size the Compton length?!?!?

What the  does this mean?

Wormhole, Time Machine



1988: Misner and Thorne show traversable wormhole solutions

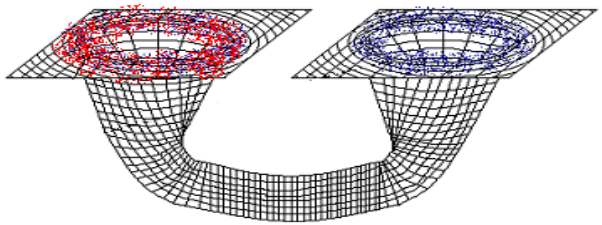
All previous solutions had event horizons and from point of view of outside observers took an infinite amount of time to traverse.

Misner and Thorne's trick: make interesting geometry and then derive the stress energy tensor consistent with it.

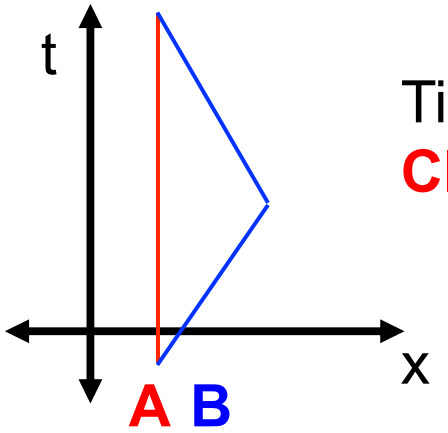
Problem: requires violation of the null energy condition
"negative energy density"

$T_{\mu\nu}k_\mu k_\nu \geq 0$ for all null vectors k_ν (NEC)

Problem: Time machines, Misner, Thorne, Yurtsever

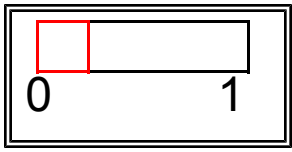


Twin paradox:



Time delay between throats.
Closed timelike curves

Dimensionality



What does gravity look like in different dimensions?

Counting argument (ADM formalism):

g_{ij} spatial metric — K_{ij} external curvature — real, symmetric, dim=d-1

degrees of freedom: $\frac{d(d-1)}{2}$ $\frac{d(d-1)}{2}$

Einstein's equation $G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi T_{\mu\nu}$

d constraints

d coordinate choices

Degree's of freedom: $\frac{d(d-1)}{2} + \frac{d(d-1)}{2} - d - d = d(d-3)$

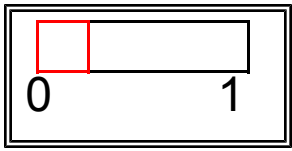
d=4 → 4 degrees of freedom

2 gravitational wave

d=3 → 0 degrees of freedom

No degrees of freedom!

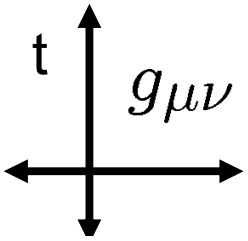
2+1 ≠ 3



Einstein vacuum equations: $G_{\mu\nu} + \Lambda g_{\mu\nu} = 0$ becomes $R_{\mu\nu} = 2\Lambda g_{\mu\nu}$

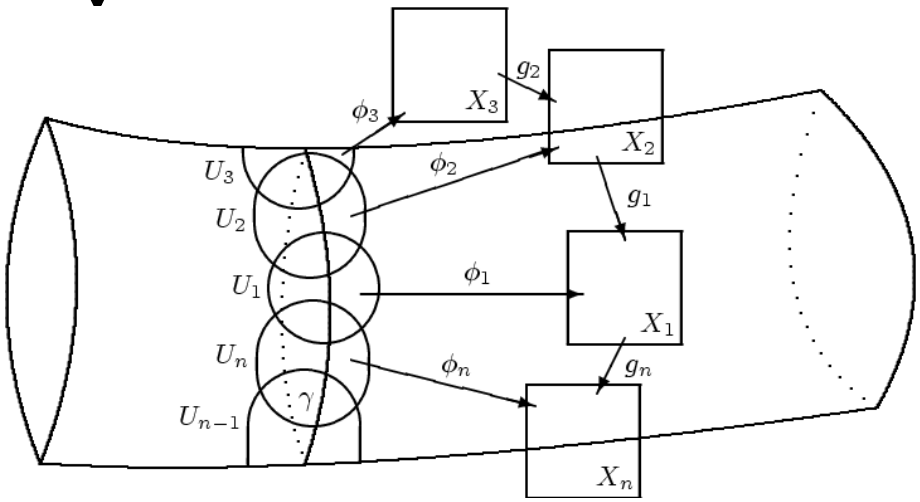
Space is everywhere flat, de Sitter or ant-de Sitter

concentrate on flat case $R_{\mu\nu} = 0$



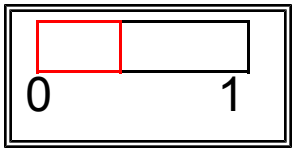
$$g_{\mu\nu} = diag(-1, 1, 1, 1)$$

Boring !

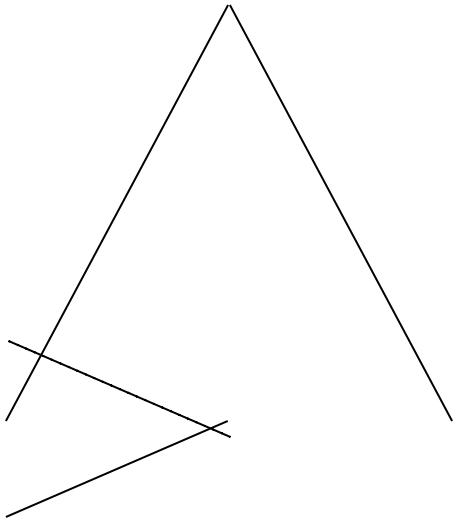


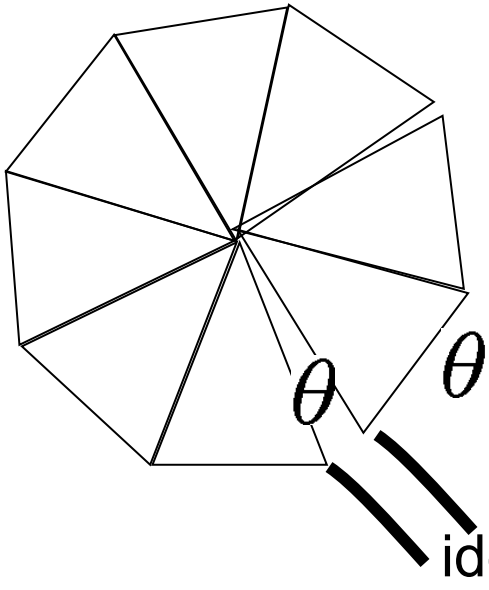
But on topologically nontrivial space, we can patch together flat space with coordinate patches. The transition functions are elements of the Poincaré group ISO(2,1)

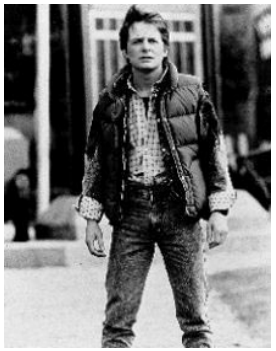
2+1 Chronology Safety?



Particles in 2+1 dimenisional gravity are conical singularities



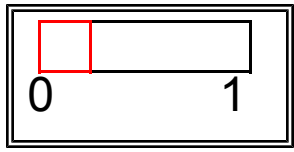




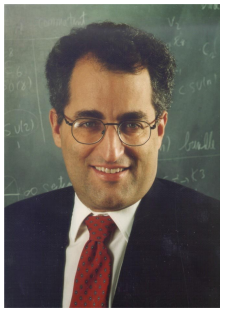
Gott constructed a spacetime in 2+1 with closed timelike curves

This set off a series of work (t'Hooft, Carroll, Farhi, Guth, Olum) showing that very generally closed timelike curves are not physical in 2+1 dimensional gravity.

Smitten with Witten



Ed Witten showed 2+1 dimensional gravity can be cast as a Chern-Simons theory with gauge group ISO(2,1)



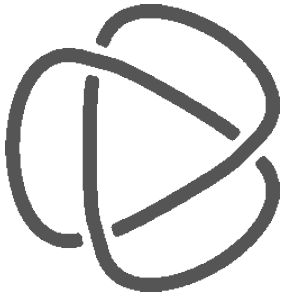
$$S_{CS}[A] = \frac{1}{8\pi G}\int_M \mathrm{Tr}\left(A \wedge dA + \frac{2}{3} A \wedge A \wedge A\right)$$

A is an ISO(2,1) valued connection

Independent of the metric of the manifold M:

Topological Quantum Field Theory (TQFT)

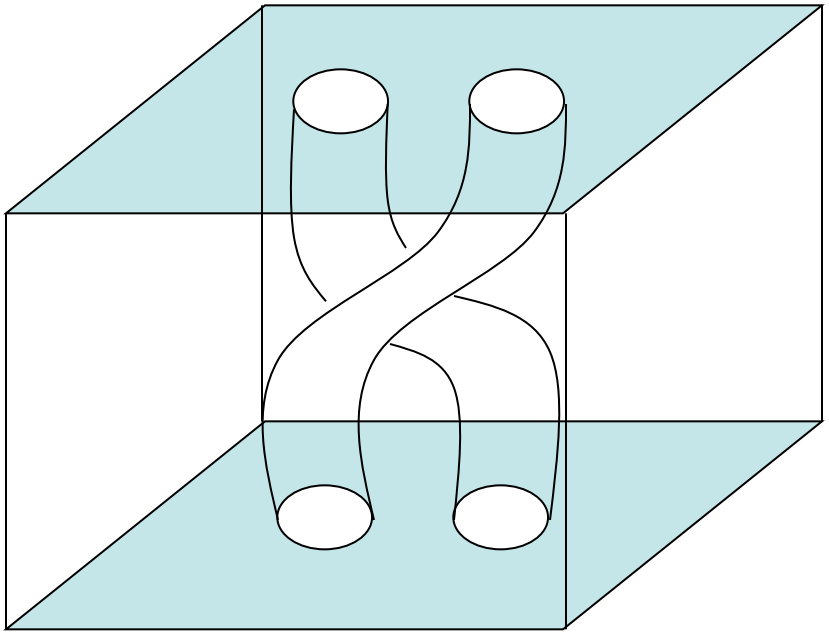
Side note: Chern-Simons theory with gauge group SU(2) leads to Wilson loops whose values are the knot invariants known as the Jones polynomial

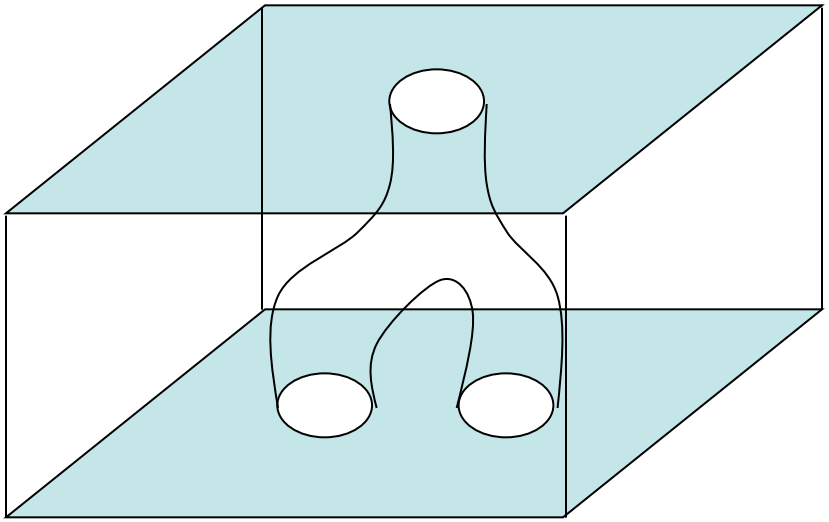


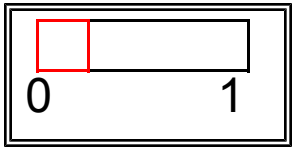
(SU(N) leads to the HOMFLY polynomial)

TQFT

"Quantum Quandries: A Category-Theoretic Perspective," J. Baez 04







Manifold M → Hilbert space H

Cobordism C → Unitary Operator U

functor

Manifold M → Hilbert space H

Manifold M → Hilbert space H

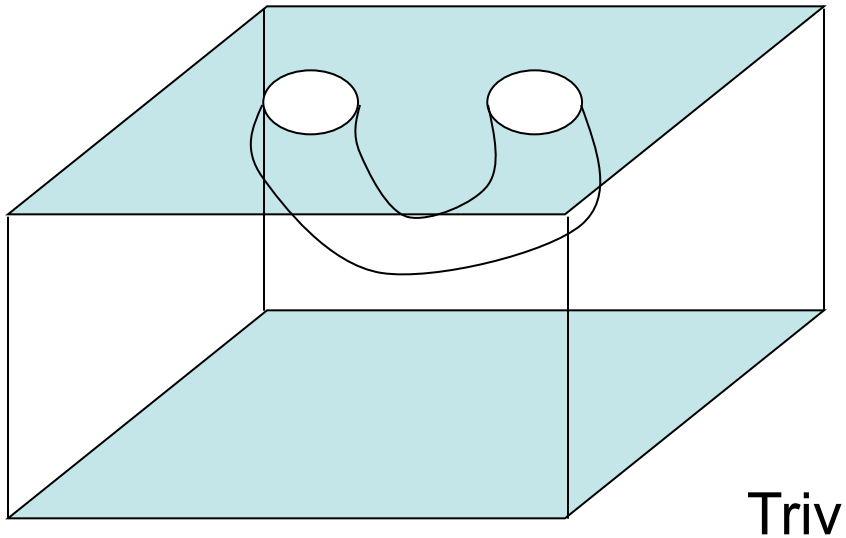
Cobordism C → Linear Operator L

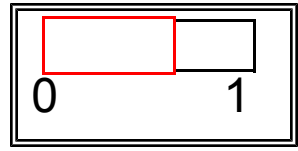
functor

Manifold M → Hilbert space H

Moral: Topology change corresponds to measurement

TQFT





Manifold M → Hilbert Space

Cobordism C → Linear Operator

functor

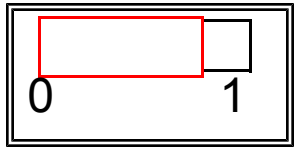
Trivial Manifold M → Hilbert space $\mathbb{C}$

We can associate preparation of a state with cobordism C from trivial manifold to a manifold of topology associated with a given Hilbert space

Realistic description of the wavefunction as cobordism from trivial.
Unitary manipulation by cobordism.
Measurement corresponds to “trimming the tree” topology chage.

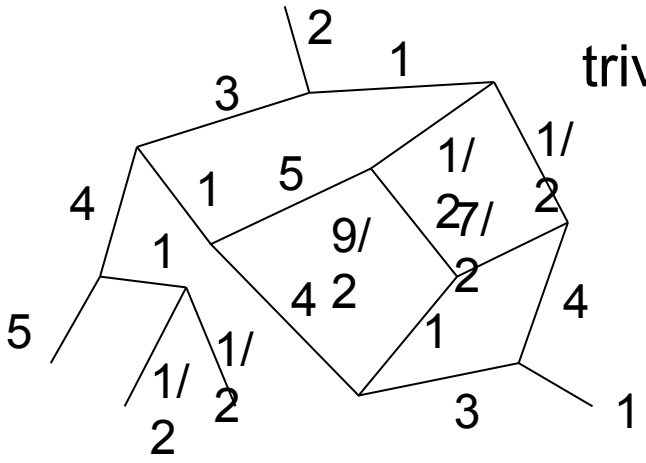
Can the topological properties allow us to get around Bell’s theorem?

But: why the quantum probability rule?



Spin Networks

Penrose's original spin networks



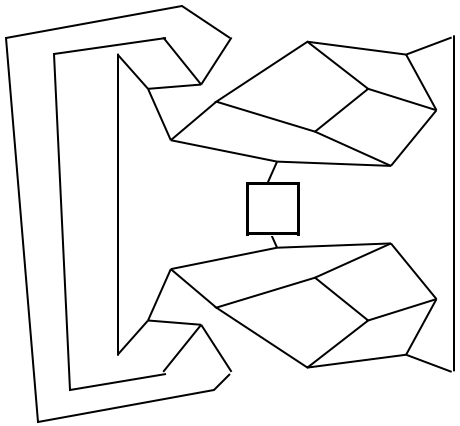
trivalent graph with edges labeled by half integers

$$l_3 = |l_1 - l_2|, |l_1 - l_2| + 1, \ldots, |l_1 + l_2|$$

+ cyclic permutations

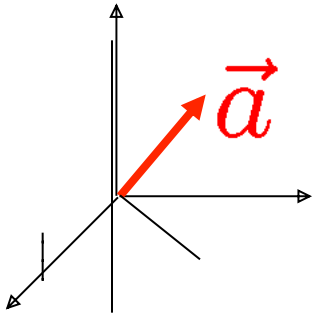
Penrose: Combinatorial rule for calculating effect of "measuring" the spin network reproduces the statistics of a single qubit.

$P_i = |\langle i|\psi\rangle|^2$ obtain rational values whose density is proportional to size of network

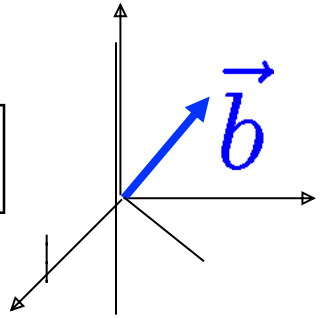


realistic description from which probabilities can be calculated is the double of the spin network.

A Protocol



$$\Pr\left[A, B|\vec{a}, \vec{b}\right] = \frac{1}{4}\left[1 - AB\vec{a}\cdot\vec{b}\right]$$



How to simulate without sending full description of vectors between Alice and Bob?

Alice and Bob share two perfectly correlated uniformly distributed unit three dimensional vectors $\vec{\lambda}_1,\ \vec{\lambda}_2$

1. Alice outpu$-\mathrm{sgn}(\vec{\lambda}_1 \cdot \vec{a})$
2. Alice sends$c = \mathrm{sgn}(\vec{\lambda}_1 \cdot \vec{a})\mathrm{sgn}(\vec{\lambda}_2 \cdot \vec{a})$ to Bob
3. Bob output$\mathrm{sgn}((\vec{\lambda}_1 + c\vec{\lambda}_2) \cdot \vec{b})$

Correlations on a single ebit
=
shared randomness + 1 bit of communication

[Bacon and Toner, *Phys. Rev. Lett.* 90, 157904 (2003)]
[Toner and Bacon, *Phys. Rev. Lett.* 91, 187904 (2003)]

Dirac Belt Tricks



Me a few years ago: Why 1 bit?
What game are quantum correlations good at?

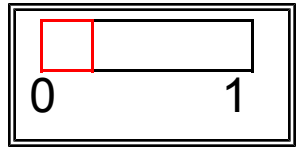
Me today: Why the geometric structure of the protocol?

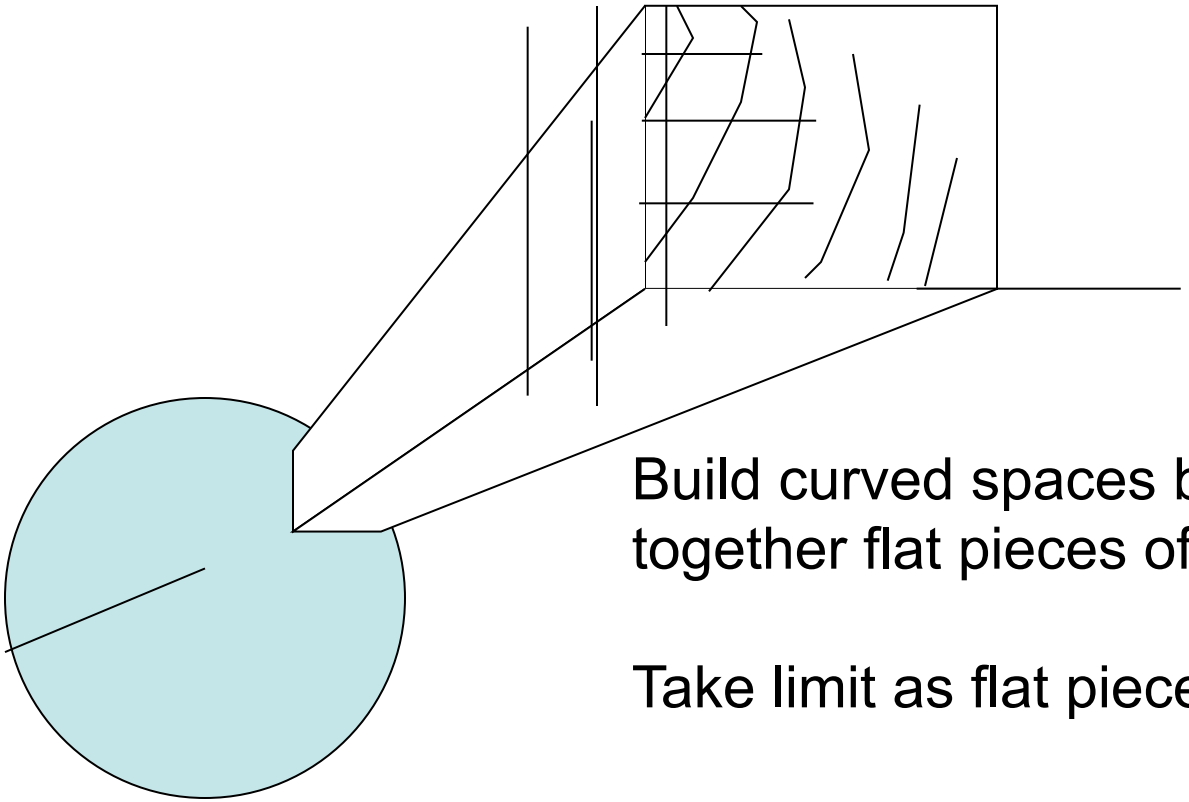


The existence of SU(2) as evidence for of particles as embedding of a geometric object in spacetime?

DEMONSTRATION

Let's Build Spacetime

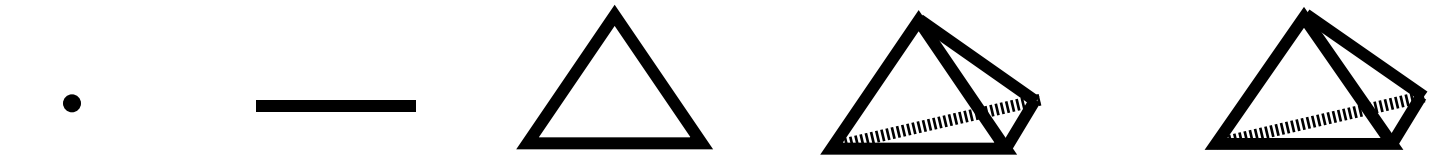




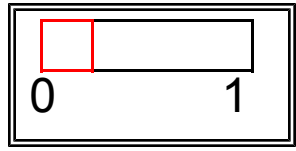
Build curved spaces by gluing together flat pieces of space.

Take limit as flat pieces shrink

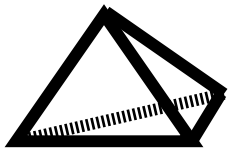
D dimensional simplexes



D=0 D=1 D=2 D=3 D=4



Regge Calculus



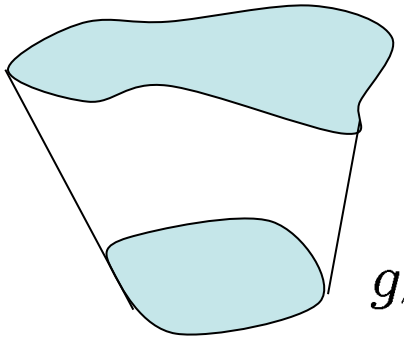
D=4

Build up spacetime by gluing together 4 dimensional flat simplexes of spacetimes with variable length edges

For given configuration can calculate the action S

Quantum Regge Calculus

Amplitude for evolution of spacetime: $\int D[g_{\mu\nu}] \exp(iS(g_{\mu\nu}))$

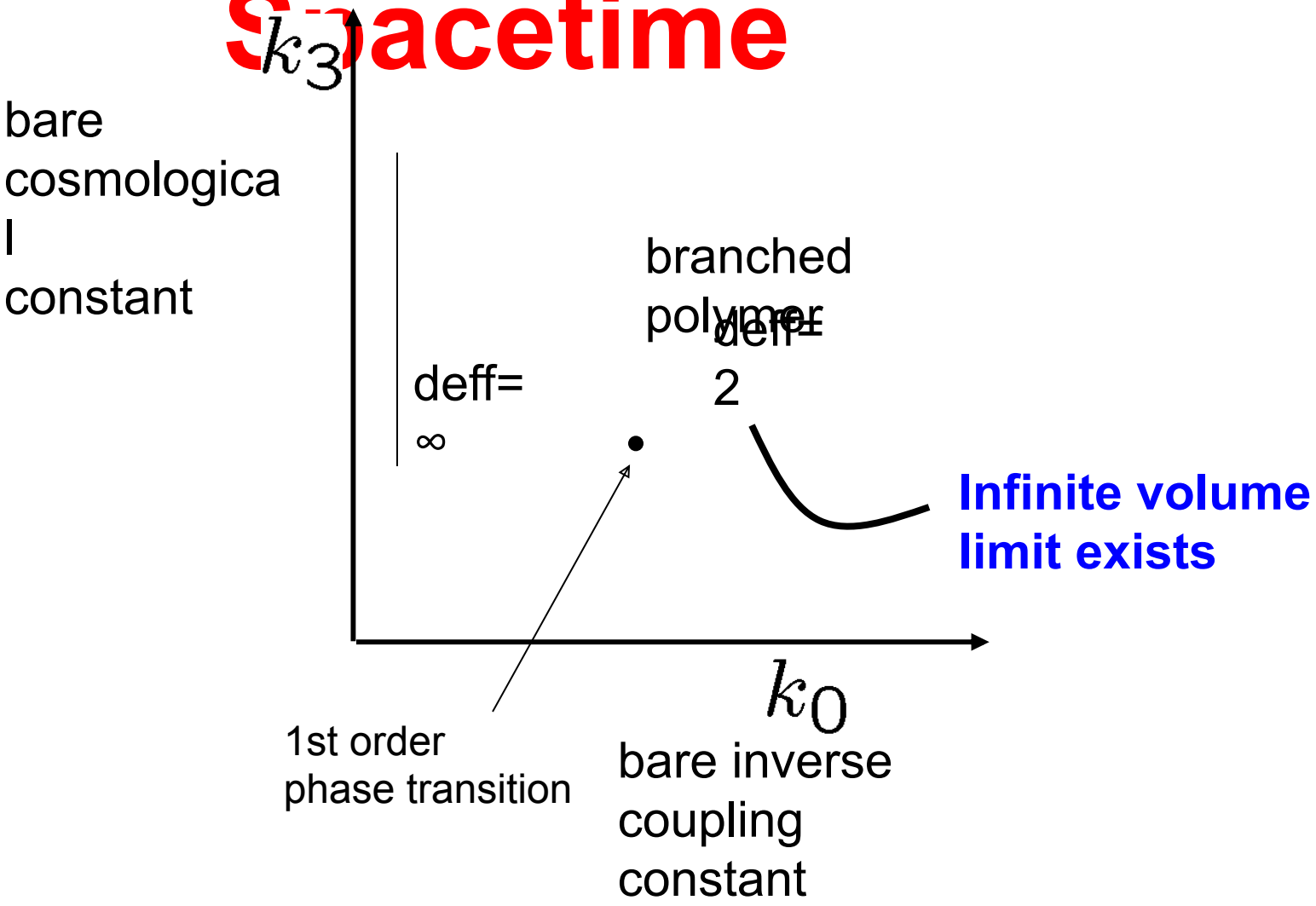


How to define this sum over geometries?

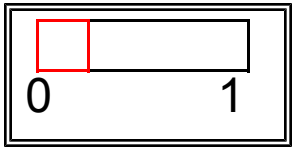
Do we include topology?
CTC spacetimes?
Singular metrics?

A common trick in the 1990s was to consider Euclidean theories and then perform a Wick rotation $\exp(iS(g_{\mu\nu})) \rightarrow \exp(-S(g_{\mu\nu}))$

Not My Spacetime



Causal Dynamical Triangulations



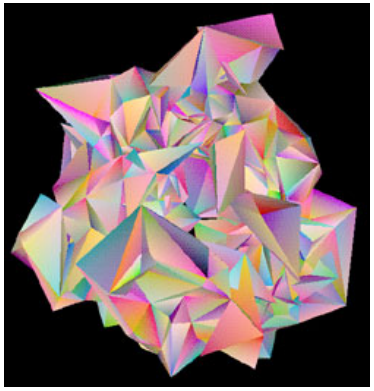
Ambjorn, Jurkiewicz, Loll, …

variation on quantum Regge Calculus

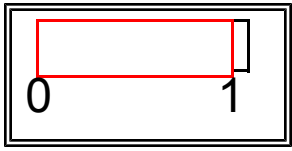
Lorentzian theories which can be rotated to Euclidean theories

spacetime topology preserving, no causality violations….

$\int D[g_{\mu\nu}] \exp(iS(g_{\mu\nu}))$ yields 4 dimensional spacetimes!



Reconciliation

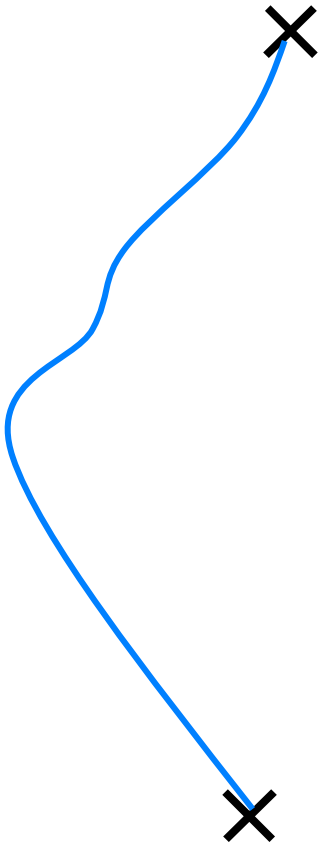


Extension of general relativity to derive quantum theory

- 3+1 nontrivial topologies problematic
- 2+1 topologies do not lead to causality problems
- 2+1 topologies lead to realistic description of wavefunction
- spin networks similarly lead to realistic description
- spin networks lead to combinatorial quantum probabilities
- half-integer spin implies relational embedding important
- dynamical triangulations give quantum theory of spacetime
- we can reinterpret dynamical triangulations as classical theory

3+1 dimensional dynamical triangulation
plus
2+1 dimensional objects which obey quantum theory

A Possible Path



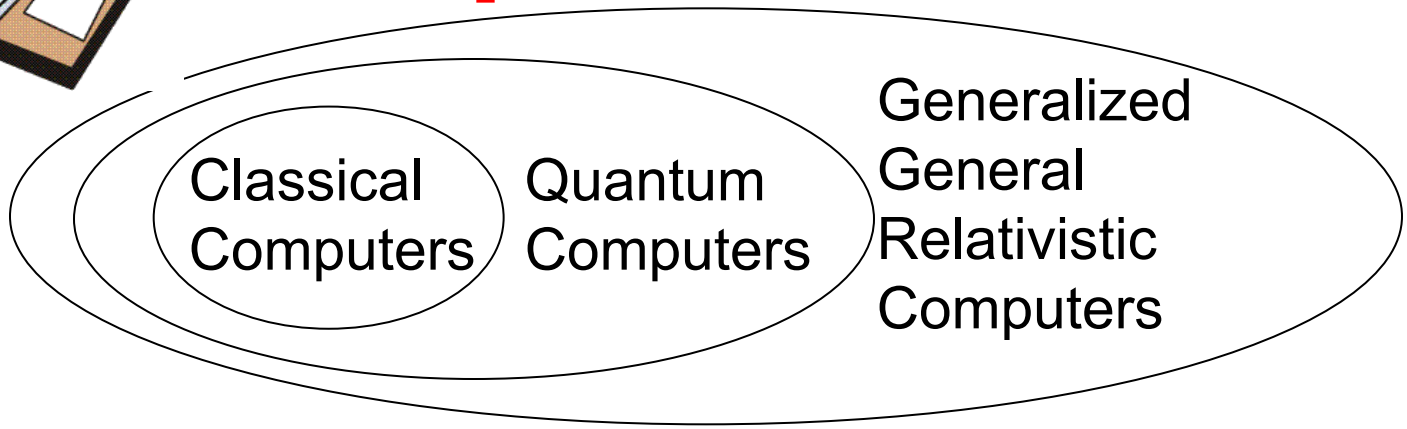
Understand how to incorporate all of the objects of the standard model into theory

Understand breaking of general covariance in merging 3+1 dimensional and 2+1 dimensional gravity

Realistic description of quantum theory from 2+1 dimensional gravity or 2+1 dimensional TQFT



Technological Implications



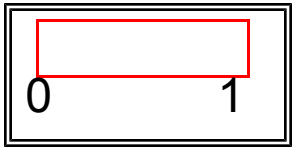
Hard to envision modifications of quantum theory which do not end up efficiently solving NP-hard problems.

Does the physical theory described here allow for the solution of NP-hard problems?

Efficiently solving NP-hard problems implies all sorts of craziness!



Technological Implications

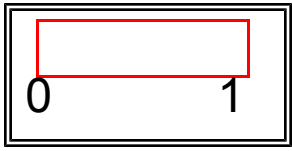


"If such a procedure existed, then we could quickly find the smallest Boolean circuits that output (say) a table of historical stock market data, or the human genome, or the complete works of Shakespeare. It seems entirely conceivable that, by analyzing these circuits, we could make an easy fortune on Wall Street, or retrace evolution, or even generate Shakespeare's 38th play. For broadly speaking, that which we can compress we can understand, and that which we can understand we can predict. Indeed, in a recent book [12], Eric Baum argues that much of what we call 'insight' or 'intelligence' simply means finding succinct representations for our sense data. On his view, the human mind is largely a bundle of hacks and heuristics for this succinct-representation problem, cobbled together over a billion years of evolution. So if we could solve the general case—if knowing something was tantamount to knowing the shortest efficient description of it—then we would be almost like gods. The NP Hardness Assumption is the belief that such power will be forever beyond our reach."

-Scott Aaronson

SIGACT News Complexity Theory Column, March 2005

Technological Implications of Being God



Polynomial Hierachy

$$\Sigma_0 P = \Delta_0 P = \Pi_0 P = P$$

$$\Delta_{i+1} P = P^{\Sigma_i P} \quad \Sigma_{i+1} P = NP^{\Sigma_i P} \quad \Pi_{i+1} P = coNP^{\Sigma_i P}$$

$$\Sigma_1 P = NP, \Delta_1 P = P, \Pi_1 P = coNP$$

$$\Sigma_2 P = NP^{NP}, \Delta_2 P = P^{NP}, \Pi_2 P = coNP^{NP}$$

Even with the power of the NP-god, there are still a series of difficult problems for us to conquer.

How far up the ladder of deity does physical reality allow?

Summary

1. Dave is Crazy

1. Quantum theory is strange



3. General relativity is rich

4. 3+1 dim gravity on nor) difficulties

5. 2+1 dim gravity has le like quantum theory

6. Put these together to yield a geometric derivation of quantum theory?

7. You should take Dave to the Looney Pen